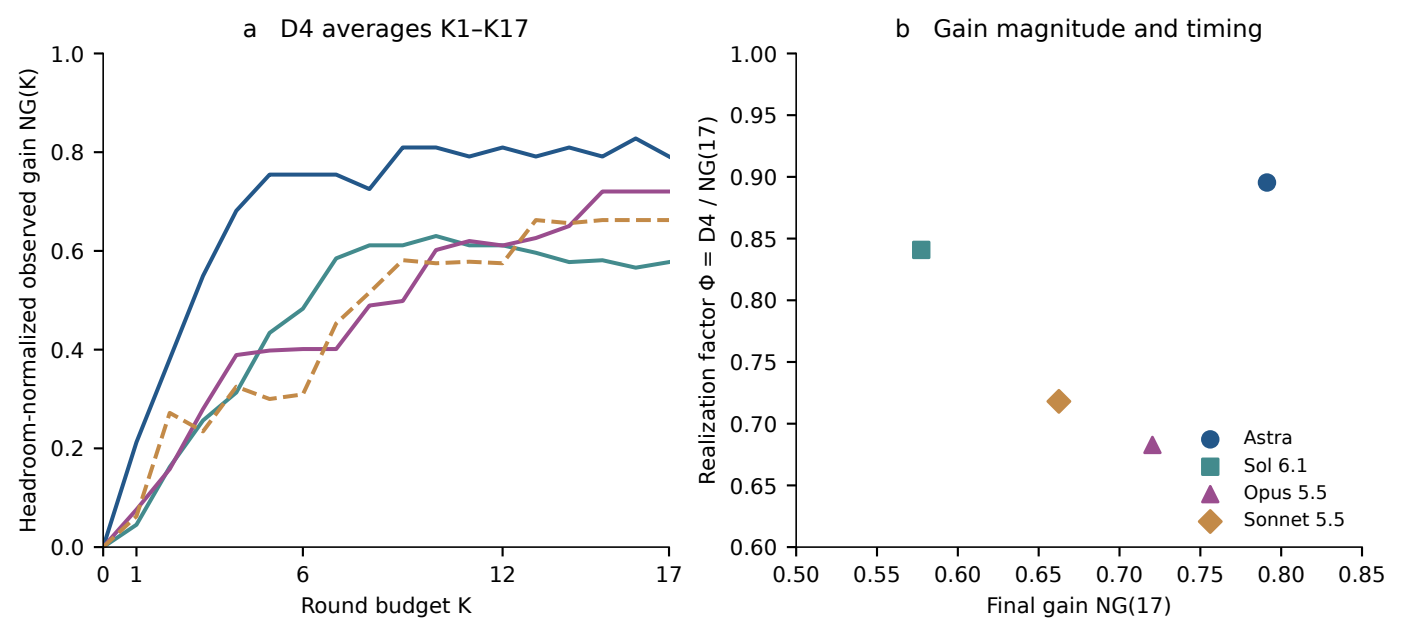
a D4 averages K1–K17
Headroom-normalized observed gain NG(K)
1.0
0.8
0.6
0.4
0.2
0.0
0
1
6
12
17
Round budget K
b Gain magnitude and timing
Realization factor Φ = D4 / NG(17)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
Final gain NG(17)
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5